MGU-BVOC (HONOURS) REGULAR EXAMINATION, NOVEMBER 2025

FIRST SEMESTER

MG1SDCAFP100-BASIC PRINCIPLES OF FOOD PROCESSING AND PRESERVATION

QP CODE: 25901014

Part- A

Short Answer

Answer any 5 questions. Each carry 2 marks

1. What are the advantages of organic foods?
    - They contain very less or no pesticides.
    - Organic foods are often fresh.
    - Organic farming is better for environment.
    - Organically raised animals are not given antibiotics, growth hormones etc.
    - They are GMO free.
2. Give examples of packing materials used in food industry?
    - Plastic containers
    - Cardboard boxes
    - Wooden crates
    - Mesh bags
    - Foam Inserts and Trays
    - Plastic Films and Bags
3. What are the principles of preservation of food?
    - Prevention or delay of microbial decomposition
    - Killing the microorganisms
    - Prevention or delay of self-decomposition of the food
    - Prevention of damage by insects, animals, mechanical causes etc.
4. Define antioxidants?
   Antioxidants are defined as "Substances that neutralize or mop up free radicals, thereby preventing or reducing oxidative stress and cell damage."
5. What do you mean by physical spoilage of food?
   Physical spoilage refers to the deterioration of food caused by physical factors such as mechanical damage, temperature fluctuations, moisture loss or gain, and improper handling or storage.
6. Write a note on energy giving function of food?

    - Provides fuel for all body activities.
    - Supports muscle work and movement.
    - Helps maintain body temperature.
    - Powers internal processes like breathing, digestion, and circulation.
    - Main energy sources: carbohydrates, fats, and proteins (to a lesser extent).

7. What do you mean by climacteric fruits?

   Climacteric fruits are fruits that continue to ripen after harvest due to a rise in respiration and ethylene production. Example- mango, banana, apple, papaya

PART-B

Short Essay

Answer any 5. Each carry 5marks.

8. What is hydrogen swell? Discuss the factors favouring hydrogen swell?
   It is the bulging or swelling of canned foods caused by hydrogen gas produced inside the can due to chemical reactions (usually between acids in food and the metal of the can).
   Factors favouring hydrogen swell:
   • High acidity of food
   • High storage temperature
   • Presence of dissolved oxygen
   • Poor quality or damaged can lining
   • Long storage time
   • Presence of metal contaminants that speed up corrosion

9. Explain fat soluble vitamins?
   Vitamins that dissolve in fats and oils, not in water.
   • They are stored in the body's fatty tissues and liver.
   • Include **Vitamins A, D, E, and K**.
   • Absorption requires dietary fat.
   Vitamin A
   • Important for vision, immunity, and healthy skin.
   • Found in liver, eggs, milk, and orange-coloured vegetables (carrots).
   Vitamin D
   • Helps in calcium absorption and bone health.
   • Produced in skin by sunlight; also in fish, egg yolk, fortified foods.
   Vitamin E
   • Acts as an antioxidant that protects cells from damage.
   • Found in nuts, seeds, and vegetable oils.
   Vitamin K
   • Essential for blood clotting and bone metabolism.
   • Found in green leafy vegetables (spinach, kale).

10. Write a note on the effect of temperature on storage of harvested products/
    - Respiration rate is decreased.
    - Water loss is reduced.
    - Ethylene production is suppressed.
    - Sensitivity to ethylene is reduced.
    - Microbial development is slowed.
11. What is the importance of functional foods?

- **Convenience:** Functional foods offer a convenient way to incorporate health-promoting ingredients into your diet.
- **Targeted benefits:** They can provide specific health benefits, such as supporting heart health or digestive health.
- **Increased nutrition:** Functional foods can enhance the nutritional value of your diet.
- **Supports overall health:** They can help prevent or manage chronic diseases, like osteoporosis or diabetes.
- **Wide availability:** Functional foods are widely available in stores and online.

12. Which are the common challenges that occur while transporting harvested products?
    - **Damage**: Rough handling, improper packaging, or long transit times can lead to bruising, crushing, or contamination of harvested products.
    - **Spoilage**: If temperature or humidity is not adequately controlled, the products may spoil during transit, resulting in financial loss and waste.
    - **Delays**: Transportation delays, due to factors like traffic, weather, or customs procedures, can lead to the deterioration of perishable goods.
    - **Cost**: Transporting perishable items, especially over long distances, can be costly due to the need for refrigerated containers, quicker transportation methods, and careful handling.
13. Define concentration and explain its types?

    Concentration is a food preservation process in which a part of the water is removed from a liquid food (like juices, milk, soups) to increase the **s**olids content and reduce volume.

    1. Heat-Based Concentration

    Uses heat to evaporate water.
    - Open pan boiling
    - Vacuum evaporation
    - Evaporators (single, double, triple effect)

    2. Membrane Concentration

    Separates water without heat using pressure and semi-permeable membranes.
    - Reverse osmosis (RO)
    - Ultrafiltration (UF)
    - Nanofiltration (NF)

    3. Freeze Concentration

    Water is removed by **freezing and ice separation** (no heating).

    4. Evaporation Assisted by Dehydration Techniques
    - Microwave evaporation
    - Thermal vapor recompression (TVR)
    - Mechanical vapor recompression (MVR)

14. Differentiate between prebiotics and probiotics?

    Prebiotics are non-digestible fibres or carbohydrates that feed the good bacteria in the gut, promoting their growth and activity.

    Probiotics are defined as: “Live microorganisms that, when administered in adequate amounts, confer a health benefit on the host."

Part- C

Essay

Answer any 1. Each carry 15marks.

15. Explain irradiation, types of rays, its advantages and disadvantages?

Irradiation:

• A food preservation method where food is exposed to ionizing radiation to kill microorganisms, insects, and delay spoilage.

Types of rays used:

• **Gamma rays**
• **X-rays**
• **Electron beams (e-beams)**

Advantages:

• Kills microbes and insects
• Extends shelf life
• Delays ripening and sprouting
• Maintains nutritional quality

Disadvantages:

• Requires expensive equipment
• Cannot kill all viruses/toxins
• Some consumers fear the process
• Not suitable for all foods (e.g., high-fat foods may get off-flavours)

16. Define food and explain its classification?

Food is defined as anything solid or liquid which when swallowed, digested, and assimilated, nourishes the body.

Classification

- Based on origin- plant and animal origin
- Based on functions- Physiological functions of food, Psychological functions of food, Social function of food.
- Based on nutrients- micro and macro nutrients
- Based on pH- low acid foods, medium acid foods, acid foods, highly acid foods